

P2P Energy Pricing Mechanism Analysis

Simulation Results and Sensitivity Analysis

Table 1: Mechanism Performance Comparison

Mechanism	Total Cost (Rs.)	Savings vs Grid	Consumer Savings	Prosumer Savings	Fairness Index	Volatility (Rs./day)
Conventional	5,743	—	—	—	1.000	11.92
MMR	5,265	8.3%	7.8%	8.8%	0.938	11.93
Bill-Sharing	5,419	5.6%	8.5%	3.2%	0.547	10.31
SDR (with DSM)	5,248	8.6%	8.1%	9.1%	0.939	12.21

Baseline: 10 households (3C + 3SP + 2MP + 2LP), grid tariff Rs.6.0/3.4, $\alpha = 12.0$, 30 days.

Table 2: Community Composition Sensitivity

Community Type	C	SP	MP	LP	MMR	Bill-Share	SDR	Best
Consumer-heavy	8	1	1	0	4.7%	4.2%	4.8%	SDR
Balanced	3	3	2	2	8.3%	5.6%	15.9%	MMR [†]
Prosumer-heavy	1	3	3	3	8.5%	4.1%	23.5%	Bill-Sharing [†]
Large PV dominated	0	1	2	7	<i>N/A (net exporter)</i>			—
Equal split (C+SP)	5	5	0	0	4.1%	3.0%	6.1%	SDR

Savings (%) vs conventional. C = consumer, SP/MP/LP = small/medium/large prosumer. [†]Best chosen by weighted score (cost 40%, fairness 30%, stability 30%), not savings alone. Net-exporting communities marked N/A as % savings is not meaningful.

Community Type	MMR	Bill-Sharing	SDR
Consumer-heavy (8C+1SP+1MP)	0.307	0.000	0.253
Balanced (3C+3SP+2MP+2LP)	0.938	0.547	0.473
Prosumer-heavy (1C+3SP+3MP+3LP)	0.973	0.906	0.468
Large PV dominated (1SP+2MP+7LP)	<i>N/A (no consumers)</i>		
Equal split (5C+5SP)	0.995	0.000	0.431

Fairness index (0–1). Measures equity of savings distribution between consumers and prosumers. N/A where only one group exists.

α	SDR Savings	Fairness	Volatility (Rs./day)	Consumer Savings	Prosumer Savings
0.05	15.9%	0.473	12.65	7.3%	23.5%
0.12	15.9%	0.473	12.65	7.3%	23.5%
0.50	15.9%	0.473	12.65	7.3%	23.5%
1.00	15.9%	0.473	12.65	7.3%	23.5%
5.00	12.2%	0.643	12.46	7.7%	16.1%
10.00	9.2%	0.860	12.25	7.8%	10.4%
30.00	6.7%	0.701	12.01	8.9%	4.8%

α = discomfort coefficient. Low $\alpha \rightarrow$ aggressive load shifting; high $\alpha \rightarrow$ DSM converges to plain SDR. Effect saturates below $\alpha \approx 1$.

Buy (Rs./kWh)	Sell (Rs./kWh)	Spread	MMR Savings	BS Savings	SDR Savings	Best
4.0	2.0	2.0	9.0%	6.1%	17.3%	MMR
6.0	3.4	2.6	8.3%	5.6%	15.9%	MMR
8.0	3.4	4.6	9.7%	6.6%	18.6%	MMR
8.0	5.0	3.0	7.6%	5.2%	14.6%	MMR
10.0	3.4	6.6	10.4%	7.1%	19.9%	MMR

Larger buy–sell spread $\rightarrow$ more room for P2P prices $\rightarrow$ higher savings for all mechanisms.

Table 3: Fairness Index by Community Type

Table 4: DSM Sensitivity (α Parameter)

Table 5: Grid Tariff Sensitivity

Table 6: Simulation Configuration

Key Observations

1. **SDR consistently achieves highest ray savings** (4.8–23.5% in net-importing communities), but fairness degrades in unbalanced compositions.

Parameter	Value	Description
Households	10	3 consumers + 3 small + 2 medium + 2 large prosumers
PV sizing (SP / MP / LP)	40–60% / 80–110% / 130–180%	Ratio of PV capacity to annual load
Grid buy price (λ_{buy})	Rs.6.0/kWh	Price households pay to import from grid
Grid sell price (λ_{sell})	Rs.3.4/kWh	Feed-in tariff for export to grid
Simulation duration	30 days	Covers weekday/weekend variation
Time resolution (Δt)	15 min (96 slots/day)	Matches smart meter granularity
DSM discomfort (α)	12.0	Quadratic penalty on load deviation
Energy conservation	Daily	Shifted load sums to same daily energy

2. **MMR provides the best fairness–savings balance** — recommended in most scenarios when fairness is weighted. Consumer and prosumer savings stay within 1–2% of each other.
3. **Bill-Sharing systematically disadvantages prosumers** (only 3.2% savings vs 8.5% for consumers), yielding poor fairness (0.547) in mixed communities.
4. **DSM effect saturates below $\alpha \approx 1$** — further reducing α yields no additional load shifting. Above $\alpha = 5$, DSM progressively disengages.
5. **Wider grid tariff spread benefits all P2P mechanisms** — the buy–sell gap defines the maximum value that P2P trading can capture.
6. **Consumer-heavy communities see lower absolute savings (4–6%)** because there is less local generation to trade.